

parameters, our $\text{bioS}(N)$ data utilizes only 3275 tokens (after applying GPT2’s tokenizer), reducing the effective embedding layer size to $3275 \times (64h)$. This adjustment explains why, for bioS data, GPT2small, typically known to have 124M parameters, is counted as having only 88M parameters in this paper.

We have selected a broad range of GPT2- ℓ - h models with practical ℓ and h values, excluding those with similar model sizes. Their selection is detailed in Figure 1, encompassing both wide and shallow transformers (e.g., GPT2-2-20, GPT2-3-20, GPT2-4-20) and skinny and deep transformers (e.g., GPT2-16-4, GPT2-16-8, GPT2-28-20). For reference, GPT2 small/med/large correspond to GPT2-12-12, GPT2-24-16, GPT2-36-20, respectively.

We primarily focus on models with $\ell \geq 2$, as 1-layer transformers may demonstrate slightly lower capacity ratios. (For those interested, 1-layer transformers are included in Figure 9, which is identical to Figure 1 but includes these models.)

Model sizes for datasets $\text{bioS}(N)$ with $N \geq 2M$. In the 1000-exposure setting, to conserve computational resources, when exploring scaling laws for $N = 2M, 5M, 10M, 20M$, we concentrate on *one model size* per dataset — specifically GPT2-16-8, GPT2-6-20, GPT2-20-16, GPT2-25-20 — as they approach the 2bit/param threshold (i.e., they satisfy $R^{\max}(F) \approx 2$). In this context, our key finding is the validation of the 2bit/param capacity ratio, thus examining a limited selection of model sizes is adequate.

For the 100-exposure setting, we evaluate a broader range of model sizes per dataset. This approach is not only due to the tenfold reduction in training time compared to the 1000-exposure setting but also to facilitate a detailed comparison of model architectures in the 100-exposure setting, aiming for precision at higher model sizes.

Training parameters. We employ the AdamW optimizer with a cosine learning rate scheduler. This includes 1K steps of warmup, followed by a cosine decay of the learning rate from 1 to 0.1 times the reference rate. We use mixed-precision fp16 training unless otherwise stated.

A.1 Base Scaling Laws

Our base scaling laws for the 1000-exposure and 100-exposure $\text{bioS}(N)$ data are presented in Figures 1(a) and 1(b), respectively.

For the 1000-exposure setting, the model’s final performance is *not very sensitive* to learning rate choices due to sufficient training. The following parameters were chosen for generating Figure 1(a):

Parameter 1 (Figure 1(a)). In the 1000-exposure setting for GPT2 models on $\text{bioS}(N)$ data:

- For $N = 10K$, we use $wd = 0.02$, $lr = 0.001$, and batch size 24 (about 140K training steps);
- For $N = 20K$, we use $wd = 0.02$, $lr = 0.001$, and batch size 48 (about 140K training steps);
- For $N = 50K$, we use $wd = 0.02$, $lr = 0.001$, and batch size 96 (about 175K training steps);
- For $N = 100K, 200K$, we use $wd = 0.02$, $lr = 0.001$, batch size 192 (about 175K, 349K training steps);
- For $N = 500K, 1M$, we use $wd = 0.01$, $lr = 0.0005$, batch size 192 (about 435K, 870K training steps);
- For $N = 2M$, we use $wd = 0.005$, $lr = 0.0003$, and batch size 1536 (about 220K training steps);
- For $N = 5M$, we use $wd = 0.002$, $lr = 0.0003$, and batch size 1536 (about 540K training steps);
- For $N = 10M$, we use $wd = 0.001$, $lr = 0.0003$, and batch size 1536 (about 1M training steps).

Remark A.1 (fp16 vs bf16). Training on GPT2 is conducted using mixed-precision fp16. We also tried bf16 and the results are nearly identical.

Remark A.2 (parameters). These optimization parameters are *very natural*, as it is generally impossible to have a fixed set of parameters for model sizes across a large multiplicative range. Notably:

- Larger model sizes naturally require smaller learning rates.

- Language models typically need at least 50K training steps *regardless of* batch size. Thus, for small N , we *reduce the batch size* to ensure the total number of training steps exceeds this threshold. For very large models, a larger batch size is preferred to enable GPU parallelism.
- When lr remains constant, wd should be relatively reduced as the number of training steps increases. Mathematically, the model weights should be “halved” for every $\Theta(\frac{1}{lr \times wd})$ training steps. Therefore, it’s advisable to reduce the wd parameter when training for longer periods.

Remark A.3 (# GPUs). In this paper, we do not specify the number of GPUs as it is irrelevant. The results remain the same whether using 64 GPUs each with a batch size of 24, 48 GPUs each with a batch size of 32, or 1536 GPUs each with a batch size of 1.

For the 100-exposure setting, careful tuning of learning rates is required. The following parameters were chosen for generating Figure 1(b): (Note: $N = 10K, 20K$ are not considered for the 100-exposure setting due to the excessively short training process.)

Parameter 2 (Figure 1(b)). In the 100-exposure setting for GPT2 models on bioS(N) data:

- For $N = 50K$, we use $wd = 0.01$, $lr = 0.001$, and batch size 12;
- For $N = 100K$, we use $wd = 0.01$, $lr = 0.001$, and batch size 24;
- For $N = 200K$, we use $wd = 0.01$, $lr = 0.001$, and batch size 48; (except for GPT2-2-20, where $lr = 0.0005$ is used)
- For $N = 500K$, we use $wd = 0.01$, $lr = 0.0005$, and batch size 96;
- For $N = 1M$, we use $wd = 0.01$, $lr = 0.0005$, and batch size 192;
- For $N = 2M$, we use $wd = 0.01$, $lr = 0.0003/0.0005/0.001$, and batch size 384;
- For $N = 5M$, we use $wd = 0.01$, $lr = 0.0003/0.0005$, and batch size 768;
- For $N = 10M$, we use $wd = 0.01$, $lr = 0.0002/0.0003/0.0005$, and batch size 1024;
- For $N = 20M$, we use $wd = 0.002$, $lr = 0.0002/0.0003/0.0005$, and batch size 1536.²²

A.2 Knowledge Memorization vs. Extraction

It was recently discovered by Allen-Zhu and Li [3] that although models memorize knowledge, this knowledge may not be extractable (e.g., via fine-tuning) for application in downstream tasks. It is essential to verify that the “2 bit/param” knowledge learned by models is indeed extractable. This verification is achieved by applying a fine-tuning task (e.g., “What is Anya’s birthday? Answer: October 2, 1996”) to half of the individuals and then testing its performance on the remainder.

Specifically, on the original bioS(N) data, we compute two quantities for each model:

- MEMORIZABLE KNOWLEDGE ACCURACY (# OF PEOPLE).

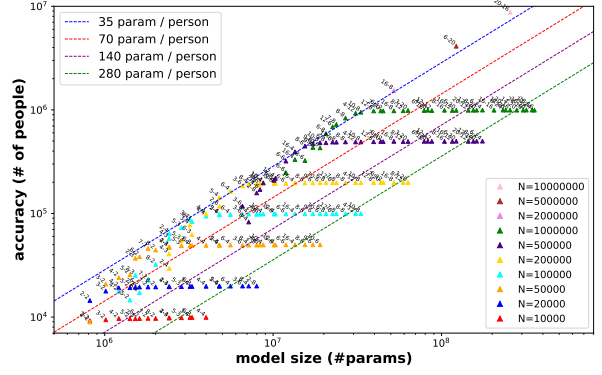
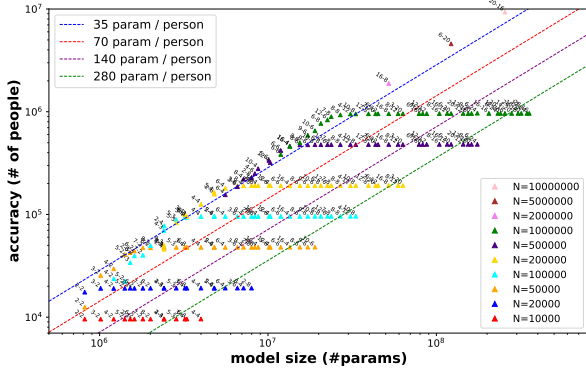
We apply the model to the original training data, such as “Anya Briar Forger was born on” and check if it can correctly generate “October 2, 1996”. For each person, we evaluate all five attributes and compute their average accuracy.²³ We then *sum* this accuracy up over all N people. (Ideally, a perfect model would have this “accuracy” equal to N .)

- EXTRACTABLE KNOWLEDGE ACCURACY (# OF PEOPLE).

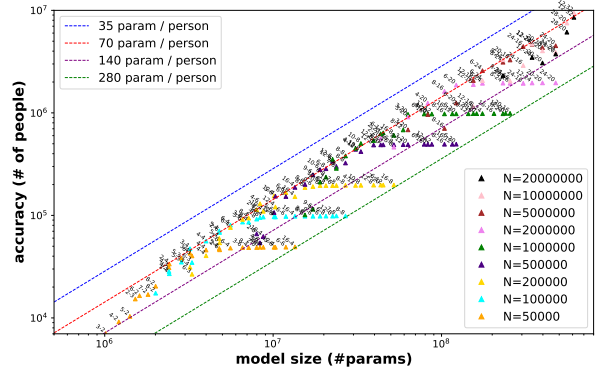
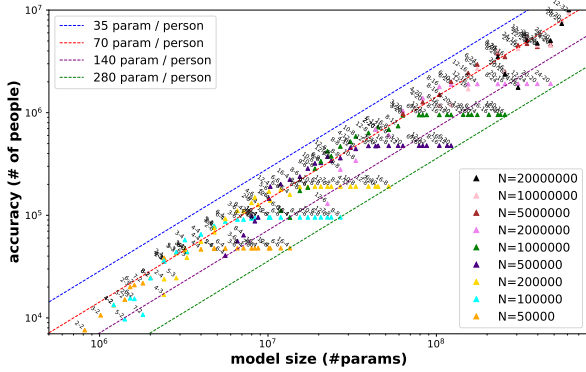
Following the pretrain-finetune framework of [3], we fine-tune any given pretrained model on half of the individuals using LoRA [17] with question-answering texts like “What is the birthday of Anya Briar Forger? Answer: October 2, 1996.” We then test its generation accuracy on the remaining half of the individuals. High accuracy indicates that the knowledge is not only memorized but can also be *flexibly* extracted for downstream tasks. Again, for each person,

²²Except for GPT2-28-20 we run out of GPU memory so reduce to batch size 1280.

²³We exclude the company city attribute because it can be uniquely determined by the employer name, thus providing no additional knowledge.



(a) **1000 exposures** — memorizable knowledge accuracy (b) **1000 exposures** — extractable knowledge accuracy



(c) **100 exposures** — memorizable knowledge accuracy (d) **100 exposures** — extractable knowledge accuracy

Figure 10: Our scaling laws from Figure 1 also apply to *extractable knowledge* (see definitions in Section A.2). This figure is for the bioS(N) datasets using GPT2 models.

Remarks. Recall from Remark 4.4 that each person’s biography contains over 47.6 bits of knowledge (excluding names), explaining why the y-axis in this figure is ~ 50 times smaller than in Figure 1.

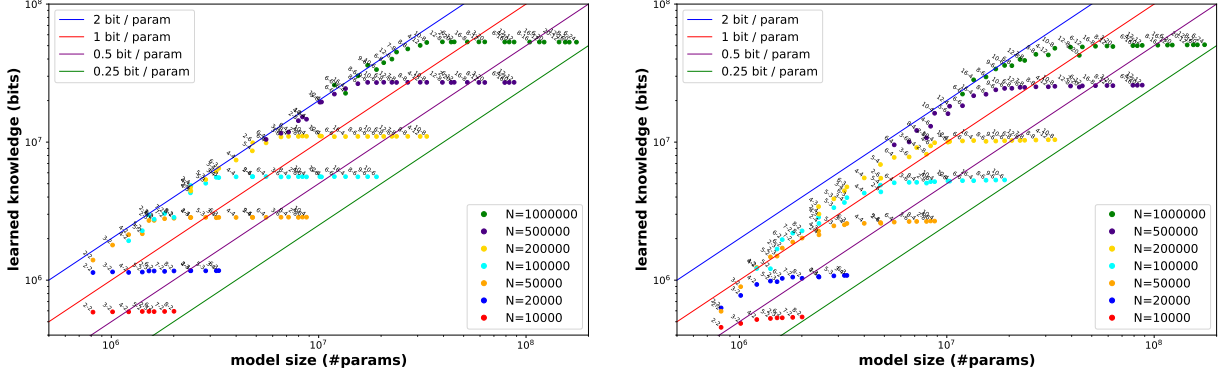
we evaluate all five attributes and compute their average accuracy. We then *sum* this across all $N/2$ people and multiply by 2. (Once again, a perfect model would have this equal to N .)

Our results are presented in Figure 10. By comparing, for instance, Figure 10(a) against Figure 10(b), it is evident that our scaling laws apply not only to memorizable knowledge but also largely to extractable knowledge. Only for models precisely at the capacity ratio boundary is there a 1.2x decrease in total accuracy.²⁴

Parameter 3 (Figure 10). When dealing with models of significantly different sizes for LoRA finetuning, it’s necessary to adjust the LoRA rank sizes. In [3], the authors primarily used a rank $r' = 128$ update for the embedding layer and ranks $r = 8$ or 16 for the query/value matrices, with their base model being either GPT2-12-12 or GPT2-12-20. In this paper, we explore a broader range of rank choices: $(r', r) \in \{(8, 2), (16, 2), (8, 4), (32, 4), (8, 8), (32, 8), (128, 8), (32, 16), (128, 16)\}$, presenting only the best results.²⁵

²⁴This decrease is in accuracy, not bits; a model may have a large amount of extractable knowledge in bits but not in accuracy. One can also compute knowledge *bits* in the extractable setting, but we omit such results for brevity.

²⁵Selecting the best LoRA option is justified as our aim is to determine the maximum extractable knowledge bits, and thus, any LoRA option demonstrating high test-set accuracy fulfills our objective.



(a) on $\text{bioS}^{\text{simple}}$ data that has no sentence diversity

(b) on the semi-real bioR data generated by LLaMA2

Figure 11: Scaling laws for the $\text{bioS}^{\text{simple}}$ and bioR data with **1000 exposures**.

We disable learning rate warmup, set the batch size to 96, the learning rate to 0.001 (with linear decay down to 0), weight decay at 0.1, and finetune for 75,000 steps.

A.3 Other Biography Datasets

We also examine the $\text{bioS}^{\text{simple}}(N)$ datasets, which are identical to $\text{bioS}(N)$ except that each individual’s knowledge is stored in a fixed ordering of six fixed sentences (see Section 2.2). Allen-Zhu and Li [3] found that in such cases, the knowledge data are memorizable but nearly 0% extractable. As shown in Figure 11(a), in these instances, the capacity ratio slightly decreases compared to Figure 1(a). This implies, in this ideal setting, adding data diversity — by rewriting the same knowledge multiple times using different writing templates — not only enhances the model’s ability to extract knowledge, as noted by [3], but also, surprisingly, *increases* the model’s capacity, as observed in this study.

Moreover, we explore the semi-real dataset $\text{bioR}(N)$, which resembles $\text{bioS}(N)$ but with the biography paragraph generated by LLaMA2, and each individual is generated 40 times (using random seeds and prompts to encourage LLaMA2 to generate as diverse paragraphs as possible for each person). This results in a total of 22GB of text, comparable to the size of Wikipedia data.

The scaling law for the $\text{bioR}(N)$ data is presented in Figure 11(b), indicating that the capacity ratio slightly decreases for larger models. This trend is expected, as LLaMA2 introduces numerous irrelevant details into the human biographies — usually different irrelevant details for each LLaMA2 generation — thereby consuming more model capacity. The decrease is more significant for smaller models, which may have greater difficulty comprehending the diverse English sentences in the data.

Parameter 4 (Figure 11). In both experiments, we adhere to the same set of optimizer parameters used in Figure 1(a), as detailed in Parameter 1.

A.4 More on Parameterized Scaling Laws

In the parameterized scaling laws, we utilize the $\text{bioD}(N, K, C, D, L, T)$ dataset from Definition 2.2.

Parameter 5 (Figure 2, 12, 13). For GPT2 models on the bioD dataset, we focus on the 1000-exposure case, with $wd = 0.01$, $lr = 0.0005$, and a batch size of 192.

Remark A.4 (parameters). Contrary to Parameter 1, it is not necessary to vary the training parameters, as our experiments with GPT2 models span a much narrower range of model sizes. We